

Electrical and Electronics Engineering

Department Knowledge Base Document

1. Department Overview

Electrical and Electronics Engineering (EEE) is a field of engineering that deals with the study and application of electricity, electronics, and electromagnetism.

The EEE Department at MLR Institute of Technology is established with state-of-the-art technical facilities to impart knowledge required for future educational and industrial needs. Key features include:

- Acoustic-proof classrooms with audio-visual teaching aids
- Departmental laboratories furnished with the latest equipment and instruments
- Internet and computer systems available across labs
- Well-experienced faculty from reputed industries and institutions
- A well-defined syllabus suited for academic, industrial, and research needs globally
- A standalone seminar hall and supporting auditorium for seminars, workshops, and training programs

The department aims to create engineers with high technical skills in EEE, suitable for employment in reputed firms and for higher studies. Key subject areas include:

- Integrated Electronics and Circuits
- Control and Automation
- Electrical Machines
- Power Electronics
- Power Systems

Students are encouraged to develop ideas that are materialized into actual projects with societal impact, and are supported to publish their work. Holistic education and professional ethics are integral parts of the curriculum. The department also focuses on overall student development through projects, seminar presentations, Student Association activities, and industrial visits.

2. Vision

To impart students with strong fundamental and applied knowledge to identify, analyze, and research upcoming energy-related challenges that provide optimal solutions to the needs of society and the nation.

3. Mission

M1: Impart core and state-of-the-art knowledge in all domains of Electrical and Electronics Engineering.

M2: Foster student learning to meet growing challenges through research and innovation.

M3: Imbibe entrepreneurial and leadership skills to develop eco-friendly energy technologies for national and societal needs.

4. Program Outcomes (POs)

The following program outcomes define the expected competencies of graduates from the EEE Department:

PO 1 – Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2 – Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3 – Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health, safety, and cultural, societal, and environmental considerations.

PO 4 – Investigation of Complex Problems: Use research-based knowledge and methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

PO 5 – Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling for complex engineering activities, with an understanding of limitations.

PO 6 – The Engineer and Society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to professional engineering practice.

PO 7 – Environment and Sustainability: Understand the impact of engineering solutions in societal and environmental contexts, and demonstrate knowledge of and need for sustainable development.

PO 8 – Ethics: Apply ethical principles and commit to professional ethics, responsibilities, and norms of engineering practice.

PO 9 – Individual and Team Work: Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary settings.

PO 10 – Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, including writing effective reports and design documentation, making effective presentations, and giving and receiving clear instructions.

PO 11 – Project Management and Finance: Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects in multidisciplinary environments.

PO 12 – Life-long Learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

5. Program Specific Outcomes (PSOs)

PSO 1: Control different Power Electronics converters and electrical machines/drives operating in different industries.

PSO 2: Provide technical solutions for different power issues related to the stability and reliability of power systems.

6. Laboratories

6.1 Power System Lab

Purpose: Study of various aspects of power systems through experiments.

Experiments conducted include:

- Measurement of transmission line characteristics data, voltage drop, and losses
- Synchronization and steady-state operation of a generator connected to an infinite bus system
- Load characteristics of a synchronous motor and effect of field excitation on reactive power load
- Effect of voltage levels on power transmission and effects of various load types on power plants
- Load flow data preparation and system study
- System analysis of symmetrical and unsymmetrical faults
- Transient stability data preparation and system study

6.2 Machines Lab (AC and DC)

Purpose: Core laboratory for basic course experiments providing versatility and generality across many curriculum needs.

This lab supports experiments on a wide variety of rotating machines and is used for research activities and energy conversion project work.

Experiments conducted include:

- Load tests on various machines
- Speed control tests
- Open circuit and short circuit tests
- Practical experiments with combinations of different rotating machines

Equipment available in this lab:

- Rectifier, motor-generator set, motor-alternator set
- Induction motor, synchronous motor
- Variety of transformers, lamp loads, inductive loads, resistive loads
- Compound machines, tachometers, voltmeters, ammeters, wattmeters
- Megger, frequency meter, synchroscope, rheostats
- Variety of starters and DC motors

6.3 EMI Lab (Instrumentation and Measurement Laboratory)

Purpose: Familiarize students with the operation of basic laboratory instrumentation and reinforce theoretical knowledge through hands-on practice.

Instruments covered include: Energy meter, multimeter, frequency counter, voltmeter, and ammeter. Students gain hands-on experience in correctly connecting circuits and using laboratory equipment.

6.4 Control Systems Lab

Purpose: Provide computing and simulation facilities and demonstrate control systems through modern equipment.

Key tools and platforms:

- MATLAB: Higher-level technical computing language for algorithm development, data analysis, and data visualization
- SIMULINK and additional toolboxes for control system design and analysis
- Real-time Workshop for real-time application implementation
- FEEDBACK-designed equipment with PCI cards for digital control system development

The lab equips students with fundamental ideas and principles underpinning modern engineering practice using versatile trainers.

6.5 Power Electronics Lab

Purpose: Enhance understanding of advanced power semiconductor devices and power electronics circuits.

This lab also provides a platform for undergraduate students to perform and test their minor projects, multi-disciplinary design projects, and major projects.

6.6 Power Semiconductor Drives Lab

Purpose: Familiarize students with the latest converters based on power semiconductor devices.

Provides basic practical knowledge in the application of power electronics in electrical drives and machines, including thyristorized speed control of DC and AC motors.

6.7 BEEE Lab (Basic Electrical and Electronics Engineering Lab)

Purpose: Provide a practical overview of theoretical concepts of basic electrical and electronics engineering.

Equipment available: Single and three-phase transformers, DC motor-generator, single-phase induction motor, and basic electronic components.

6.8 ECA Lab (Electrical Circuits and Analysis Lab)

Purpose: Expose students to electrical circuits and build circuit construction skills using different circuit elements.

7. Computer-Based Skills

The EEE Department integrates computer and coding skills into the curriculum through the following initiatives:

- Inclusion of soft computing techniques as a professional elective course (under MLR 20 regulations) to build awareness of artificial intelligence
- Open elective courses offered: Core Java Programming, Data Analytics, Machine Learning, Cloud Computing, Big Data
- Guidance provided for mini and micro projects implementing interdisciplinary concepts such as IoT and smart technologies

- Support for online internships on computer coding platforms and continuous training through Campus Recruitment Training
- Faculty expertise in artificial intelligence to guide students in implementing AI controllers in electrical engineering applications
- Short-term training programs and hackathons on Artificial Intelligence, Cyber Security, and Machine Learning provided by the Centre of Innovation and Entrepreneurship

8. Department Achievements

8.1 Interconnection of Grid and Solar Power at MLRIT Campus

MLR Institute of Technology (an autonomous institution) set up a rooftop solar (RTS) photovoltaic plant with a capacity of 260 KWp. Key details:

- Installed on two shadow-free building rooftops, covering approximately 780 sq. m.
- Uses 260 KWp polycrystalline silicon solar panels
- Connected to the power grid of Telangana State Southern Power Distribution Company Limited (TSSPDCL)
- Only load exceeding plant capacity is drawn from the electricity board, reducing energy consumption
- Annual energy generation estimated at 3,79,600 units, with approximately 10% surplus returned to TSSPDCL
- Total project cost: Rs. 1.5 crore; operational since 2018
- Part of the National Action Plan on Climate Change for clean energy promotion

8.2 Street Light Automation at MLRIT Campus

The Department of EEE successfully completed and installed the Street Light Automation Project at MLRIT Campus. The project has been operational since 18/02/2020.

Three modes of control are provided: Time Control, Remote Control, and Manual Control.

Funding: Rs. 19,000/- provided by MLRIT

Project Team:

- A. Shubhangi Rao, Asst. Prof., MLRIT – Project Coordinator
- M. Srinivas Reddy, Assoc. Prof., MLRIT
- P. Sailesh, Student, 3rd Year
- Rakshanda, Student, 3rd Year
- Sukumar, Student, 2nd Year
- Sathvik, Student, 2nd Year

8.3 Guest Lecture and Industrial Visits

- Guest lecture by Dr. Y. Pradeep Kumar, IIIT Hyderabad, on Campus Energy Monitoring System – 23/01/2020
- MLR17 batch industrial visit to 33/11 kV substation, Dundigal – 01/11/2019

9. Student Achievements

9.1 Academic and Technical Competitions

Student Name	Achievement	Organization	Year
P. Sailesh and V. Rakshanda	1st Position – Inter-Departmental Micro Project Competition	MLRIT, Hyderabad	2017 and 2018
G. Vikshit and TVS Madhur	1st Position – Departmental Micro Project Competition	MLRIT, Hyderabad	2019
P. Sailesh	Student Mentor Award – INVENTRON Event	MLRIT	02/11/2019
Ch. Sai Prakash	Quarterfinalist – AICTE, DST and Texas India Innovation Challenge Design Contest	IIM Bangalore	2019

9.2 Sports Achievements

Student Name	Achievement	Organization	Year
Vishnu Mukherjee (17R21A0260)	1st Position – Football	MLRIT, Hyderabad	2017 and 2018
Mihir Srivastava (17R21A0233)	1st Position – Football	MLRIT, Hyderabad	2017 and 2018
Sindhu P (18R21A0246)	RFYS 2K18 Athletics Championship	Reliance Foundation for Youth Sports, Gachibowli	2018
Sindhu P (18R21A0246)	Golden Mile Run	Athletics Coaching Academy, Telangana Athletics Association and Hyderabad District Athletics Association, Gachibowli	2018
Sindhu P (18R21A0246)	University Selections – BITS, VIGNAN, MLRITFEST, BVRITW, VNRVJIET	JNTUH, Hyderabad	2018
Sindhu P (18R21A0246)	Spardha 2K19	IIT BHU, Varanasi	2018
Sindhu P (18R21A0246)	NITW Sports Fest	NIT Warangal	2018
Ujwala Y (18R21A0259)	RFYS 2K18 Athletics Championship	Reliance Foundation for Youth Sports, Gachibowli	2018
Ujwala Y (18R21A0259)	Golden Mile Run	Athletics Coaching Academy, Telangana Athletics Association, Gachibowli	2018

Ujwala Y (18R21A0259)	University Selections – BITS, VIGNAN, MLRITFEST, BVRITW, VNRVJIET	JNTUH, Hyderabad	2018
Ujwala Y (18R21A0259)	Spardha 2K19	IIT BHU, Varanasi	2018
Ujwala Y (18R21A0259)	NITW Sports Fest	NIT Warangal	2018
Shaik Rasheed (18R21A0245)	Semi-Final Winner – Basketball, University Selections	JNTUH, Hyderabad	2018
Shaik Rasheed (18R21A0245)	1st Position – Basketball, University Selections	JNTUH, Hyderabad	2018

9.3 Internships

- P. Supriya Reddy, Karthik Sai Nadikudi, R. Sai Kumar, Ch. Kaveri, R. Sai Varun, V. Soumya, M. Alaikya, P. Jaswanthi, TVS Madhur, B. Kavyanjali, B. Vani – Interned at BHEL (Study of Turbo Generators)
- Karthik Sai Nadikudi, P. Jaswanthi, and Musham Gnanendra Saikumar – Interned at KTPS (Study of Power Generation)
- P. Sailesh – Interned at MLRIT (Product Development Bootcamp, IoT)

10. Faculty Achievements

Faculty Name	Achievement / Activity	Year
A. Shubhangi Rao (Asst. Prof., EEE)	Attended 3-day IUCEE Workshop	Dec 2019
A. Shubhangi Rao (Asst. Prof., EEE)	Interned at MLRIT – Product Development Bootcamp, IoT	June 2019 and June 2018
A. Shubhangi Rao (Asst. Prof., EEE)	Attended IUCEE Workshop on Design Thinking	June 2019 and June 2018
N. Karthik (Asst. Prof., EEE)	Attended 3-day IUCEE Workshop	Dec 2019
M. Sreeniwas Reddy (Assoc. Prof., EEE)	Presented paper: 'Electrical Power Distribution System Fault Location and Rectification Using Modern Technologies to Universities and Multistoried Buildings' at International Conference (IIAR)	–
M. Sreeniwas Reddy (Assoc. Prof., EEE)	Presented paper: 'Protection of Power System Components for Renewable Energy Sources' at International Conference (IIAR)	21 Jan 2018
M. Sreeniwas Reddy (Assoc. Prof., EEE)	Attended FDP on 'IoT Applications in Green Energy Systems'	23 Apr – 3 May 2018

11. Publications

11.1 Faculty Publications

S.No	Faculty Name	Title of Paper	Journal	Year
1	Dr. A. Sudhakar	Harmonic Minimization in Direct Torque Controlled Induction Motor using Neural Network Controller	International Journal of Energy Technology and Policy (Scopus indexed)	2020
2	M. Srinivasa Reddy	MLR Institute of Technology Campus Energy Monitoring and Controlling System with Interconnection of Grid and Solar Power	IJET, UAE (SCOPUS)	Apr 2020
3	M. Srinivasa Reddy	Mitigating Voltage Sag in Secondary Distribution Line Using DVR with Single DC Source	International Journal of Scientific and Technology Research, Vol. 9, Issue 03	Mar 2020
4	M. Srinivasa Reddy	Review on Energy Consumption by Using Solar Systems	(IJPAM), Vol-120, No. 6, ISSN: 1314-3395	Oct 2018
5	M. Srinivasa Reddy	Design of Sensors for Adaptive Solar Panels	ARPN Journal of Engineering and Applied Sciences, Vol-13, No.10, ISSN: 1819-6608	May 2018
6	M. Srinivasa Reddy	Automatic Car Washing System with IR Module using Conveyer Belt	(IJSDR), Vol-2, Issue-5, ISSN: 2455-2631	May 2017
7	M. Srinivasa Reddy	New Hybrid PSO Algorithm for Non-Convex Optimal Power Flow Problem	(IJEER), VOL-4, Issue-4, ISSN: 2348-6988	Dec 2016
8	M. Srinivasa Reddy	Solar Electric Battery Car using Microcontroller	(IJEER), Vol-4, Issue-2, ISSN: 2348-6988	Apr 2016
9	M. Srinivasa Reddy	Electric Battery Car using Microcontroller	(IJARET), Vol-3, Issue-IV, ISSN: 2320-6802	Jun 2015
10	M. Srinivasa Reddy	An Intelligent RFID Reader and its Application for Airport Baggage Tracking System	(IJGIS), Vol-1, Issue-2, ISSN: 2319-9245	Oct 2014
11	M. Srinivasa Reddy	Power Factor Correction	(IJIES), ISSN: 2319-9598, VOL-1, Issue-12	Nov 2013
12	M. Srinivasa Reddy	Digital Computation of Fault Location on Electric Power Transmission Lines	(IJERIA), ISSN: 0974-1518, VOL-3, No. 3	Aug 2010
13	T. Bhargava Ram	Mitigating Voltage Sag in Secondary Distribution Line Using DVR with Single DC Source	International Journal of Scientific and Technology Research, Vol. 9, Issue 03	Mar 2020

14	T. Bhargava Ram	Investigation in Finding the Role of Active Loads in the Electrical Grid Dominated by Power Electronics	Think India Journal	Dec 2019
15	T. Bhargava Ram	Performance Analysis of Different Level D-Statcom using Finfet Technology	History Research Journal	Aug 2016
16	T. Bhargava Ram	BLDC Motor Based CUK Converter using Model Predictive Controller for Low Power Applications	International Journal of Advanced Trends in Computer Science and Engineering	Published
17	Ch. Srivardhan Kumar	Mitigating Voltage Sag in Secondary Distribution Line Using DVR with Single DC Source	International Journal of Scientific and Technology Research, Vol. 9, Issue 03	Mar 2020
18	Ch. Srivardhan Kumar	Mitigation of Harmonics and Enhancement of Power Quality with Solar Based DVR System	International Journal of Pure and Applied Mathematics	2018
19	Ch. Srivardhan Kumar	Review on Energy Consumption by Using Solar Systems	International Journal of Pure and Applied Mathematics	2018
20	Ch. Srivardhan Kumar	BLDC Motor Based CUK Converter using Model Predictive Controller for Low Power Application	International Journal of Advanced Trends in Computer Science and Engineering	Published
21	A. Shubhangi Rao	MLR Institute of Technology Campus Energy Monitoring and Controlling System with Interconnection of Grid and Solar Power	IJET, UAE (SCOPUS)	Apr 2020
22	A. Shubhangi Rao	Energy Management System on Operation of Smart Grid	IRJET	Mar 2020
23	A. Shubhangi Rao	BLDC Motor Based CUK Converter using Model Predictive Controller for Low Power Application	International Journal of Advanced Trends in Computer Science and Engineering	Published
24	Jitendar	Minimization of Power Disturbance by Using EVP for Critical Load Appliances	–	–
25	Jitendar	Enhancement of Steady State Performance in 1-Phase Rectifier for High Power Application using Active Power Decoupling Method	–	–

11.2 Student Publications

Student: Sai Prakash, III Year

- Energy Management System through Operation of Smart Grid, IRJET, 2020
- MLR Institute of Technology Campus Energy Monitoring System through Interconnection of Grid and Solar Power, IJET, 2020